T300 | Carbon Fiber Data Sheet

Toray Carbon Fibers Europe

A carbon fiber industry benchmark with a 50-year production history. Torayca™ T300 is renowned for its balanced composite properties, visual finish, high quality, consistency and reliability. Excellent processability in traditional manufacturing methods (weaving, braiding, filament winding, prepreg, etc). Manufactured in France and Japan.

PRODUCT DESIGNATION

T300	B	3 000	4	0	A
Fiber Type	Twist*	Filament Count	Sizing Type	Surface Treatment	Sizing Amount

*Blank: Twisted
B: Untwisted

RESIN SYSTEM COMPATIBILITY

SIZING TYPE & AMOUNT	RESIN SYSTEM COMPATIBILITY	METHOD
40A / 40B (1%)	Epoxy, phenolic, BMI	TY-QA-005
40D (0.7%)	Epoxy, phenolic, BMI	TY-QA-005
50A / 50B (1%)	Epoxy, phenolic, polyester, vinyl ester	TY-QA-005

FIBER PROPERTIES

PROPERTY	UNIT	FILAMENT COUNT	NOMINAL VALUE*	METHOD
Tensile Strength	MPa (kgf/mm²)		3530 (360)	TY-QA-002
Tensile Modulus	GPa (10^3kgf/mm²)		230 (23.5)	TY-QA-002
Elongation	%		1.5	TY-QA-002
Density	g/cm³		1.76	TY-QA-003
Yield	g/1000m	1 000	66	TY-QA-004
		3 000	198	
		6 000	396	

* Lot properties are reported as average values of a test set per bobbin sampled. The stated values are typical values. For design purposes, please contact us.

FUNCTIONAL PROPERTIES

PROPERTY	UNIT	FILAMENT COUNT	NOMINAL VALUE
Specific Heat	Cal/g.°C		0.19
Electric Resistivity	x 10^{-3}Ω.cm		1.7
CTE	α10^{-6}/°C		-0.41
Thermal Conductivity	Cal/cm.S.°C		0.025
Filament Diameter	µm		7
Cross Sectional Area	mm²	1 000	0.04
		3 000	0.11
		6 000	0.23

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COMPOSITE PROPERTIES*

PROPERTY	UNIT	NOMINAL VALUE
Tensile Strength	MPa (kgf/mm²)	1760 (180)
Tensile Modulus	GPa (10^3kgf/mm²)	132 (13.5)
Tensile Strain	%	1.3
Compressive Strength	MPa (kgf/mm²)	1570 (160)
Flexural Modulus	GPa (10^3kgf/mm²)	125 (13.0)
ILSS	MPa (kgf/mm²)	110 (11)

* Toray 3631 – 180°C resin system. Measured temperature: RT. Normalized to 60% fiber volume.

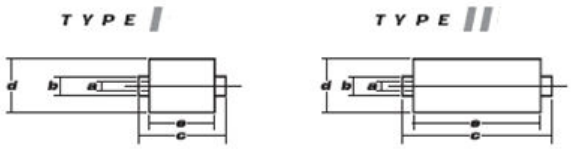
PACKAGING

Each bobbin of TORAYCA carbon fibre is protected against dust and packed in container to prevent damage during transportation

The table below summarizes the tow sizes and packaging available for standard material. Other bobbin sizes may be available on a limited basis.

TOW SIZES	BOBBIN NET WEIGHT (kg)	BOBBIN TYPE	BOBBIN SIZE (mm)					SPOOL PER CASE	CASE NET WEIGHT (kg)
			a	b	c	d	e		
1K	2.0	I	76	82	192	132	156	16	16
3K	2.0	I	76	82	192	157	156	12	24
	4.0	I	76	82	192	210	156	150	600
6K	2.0	I	76	82	192	157	156	12	24

BOBBIN TYPE:



Please refer to SDS for handling and disposal.

REQUIREMENTS

Shelf Life: 24 months from month of manufacture, if stored under ambient condition

TORAY CARBON FIBERS EUROPE S.A.

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